

SIMATS ENGINEERING

**SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
CHENNAI – 602105**

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – APPLICATION BASED QUESTIONS

Course Code:	<u>DSA0501</u>
Course Title:	<u>QUERY PROCESSING</u>
CO Assessed:	CO1
Assessment:	Tool 2 – Application Based Questions
Weightage:	_____
Total Marks:	20 Marks

CO1 Statement: _____

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Section/Batch: A-2025_ Date of Submission: 27-07-26

Faculty In-charge: VEENA_____ Submission Mode: Individual

Code of Conduct

I DENISHA SHREE R_ certify that this submission is my original work and have adhered to all academic integrity guidelines.

DENISHA SHREE
Signature:

Criteria	Marks
Understanding of the Problem	4
Application of Concepts	4
Program Design / Algorithm	4
Correctness of Solution & Output	4
Explanation / Interpretation of Results	4
Total	20

O1 – AT2 – Application Based Questions

Sample Questions

1.College Attendance Management (10 Marks)

A college maintains daily attendance records in a CSV file. The file contains duplicate attendance entries, missing roll numbers, and inconsistent date formats.

Write a Python program to read the file, clean the dataset, remove duplicate records, standardize the dates, and generate a new CSV file that can be used for attendance analysis.

PYTHON CODE:

```
import pandas as pd

# Read the CSV file
df = pd.read_csv("attendance.csv")

# Display original data
print("Original Dataset:")
print(df)

# -----

# Data Cleaning
# -----

# Remove duplicate records
df = df.drop_duplicates()

# Remove rows with missing Roll Numbers
df = df.dropna(subset=["Roll_Number"])

# Standardize Date format (YYYY-MM-DD)
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df = df.dropna(subset=["Date"])
df["Date"] = df["Date"].dt.strftime("%Y-%m-%d")

# Save cleaned dataset
df.to_csv("attendance_cleaned.csv", index=False)

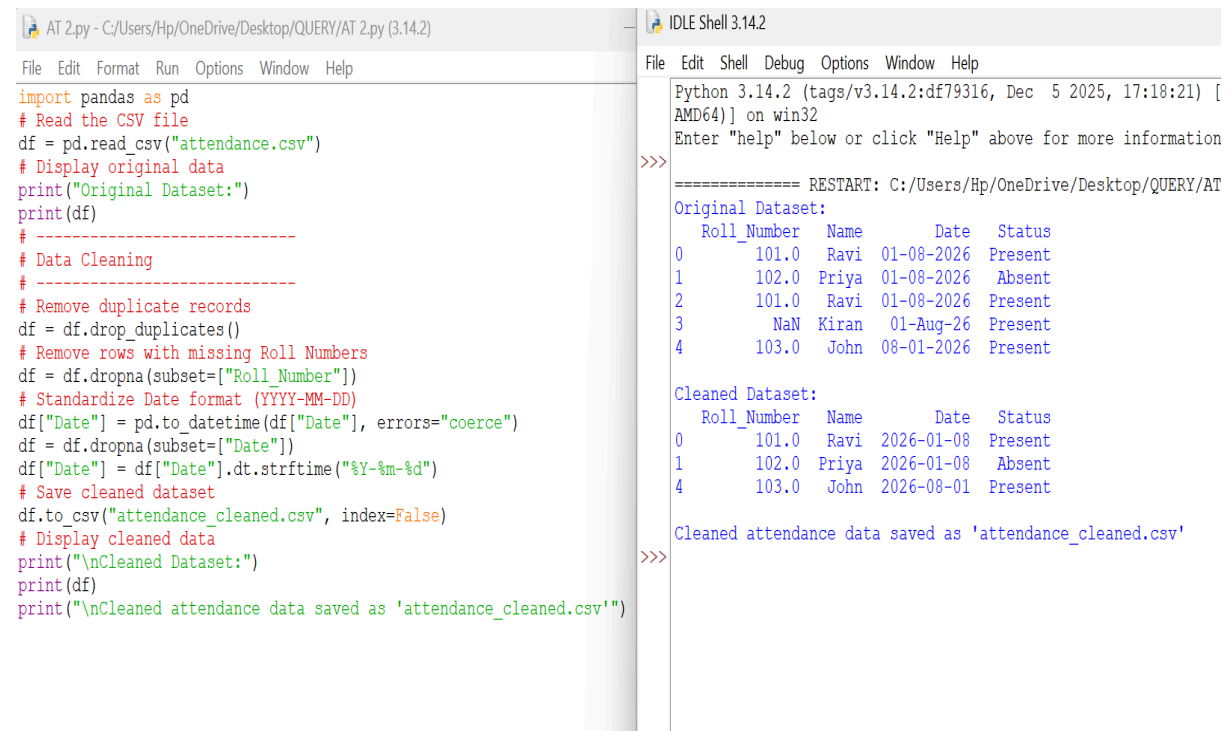
# Display cleaned data
```

```
print("\nCleaned Dataset:")
```

```
print(df)
```

```
print("\nCleaned attendance data saved as 'attendance_cleaned.csv'")
```

OUTPUT:



The screenshot shows a Python IDE with two panes. The left pane contains the source code for a script named 'AT 2.py'. The code imports pandas, reads a CSV file 'attendance.csv', displays the original data, performs data cleaning (removing duplicates and rows with missing roll numbers), standardizes the date format, and saves the cleaned data to 'attendance_cleaned.csv'. The right pane shows the output of the script, which includes a restart message, the original dataset as a table, the cleaned dataset as a table, and a confirmation message that the cleaned data has been saved.

```
AT 2.py - C:/Users/Hp/OneDrive/Desktop/QUERY/AT 2.py (3.14.2)
File Edit Format Run Options Window Help

import pandas as pd
# Read the CSV file
df = pd.read_csv("attendance.csv")
# Display original data
print("Original Dataset:")
print(df)
# -----
# Data Cleaning
# -----
# Remove duplicate records
df = df.drop_duplicates()
# Remove rows with missing Roll Numbers
df = df.dropna(subset=["Roll_Number"])
# Standardize Date format (YYYY-MM-DD)
df["Date"] = pd.to_datetime(df["Date"], errors="coerce")
df = df.dropna(subset=["Date"])
df["Date"] = df["Date"].dt.strftime("%Y-%m-%d")
# Save cleaned dataset
df.to_csv("attendance_cleaned.csv", index=False)
# Display cleaned data
print("\nCleaned Dataset:")
print(df)
print("\nCleaned attendance data saved as 'attendance_cleaned.csv'")

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [AMD64] on win32
Enter "help" below or click "Help" above for more information
>>>
===== RESTART: C:/Users/Hp/OneDrive/Desktop/QUERY/AT
Original Dataset:
   Roll_Number  Name      Date  Status
0         101.0   Ravi  01-08-2026  Present
1         102.0  Priya  01-08-2026   Absent
2         101.0   Ravi  01-08-2026  Present
3           NaN  Kiran  01-Aug-26   Present
4         103.0   John  08-01-2026  Present

Cleaned Dataset:
   Roll_Number  Name      Date  Status
0         101.0   Ravi  2026-01-08  Present
1         102.0  Priya  2026-01-08   Absent
4         103.0   John  2026-08-01  Present

Cleaned attendance data saved as 'attendance_cleaned.csv'
>>>
```

2.Electricity Consumption Analysis (10 Marks)

An electricity board receives monthly power consumption data from different cities in CSV format. Some records contain invalid meter readings and missing consumer IDs. Write a Python program to identify incorrect records, clean the dataset, and generate a summary showing the total electricity consumption for each city.

PYTHON CODE:

```
import pandas as pd
```

```
# Read CSV file
```

```
df = pd.read_csv("electricity.csv")
```

```
print("Original Dataset:")
```

```
print(df)
```

```
# -----
```

```
# Data Cleaning
```

```
# -----
```

```
# Remove rows with missing Consumer ID
```

```

df = df.dropna(subset=["Consumer_ID"])

# Convert Meter_Reading to numeric
df["Meter_Reading"] = pd.to_numeric(df["Meter_Reading"], errors="coerce")

# Remove invalid meter readings (negative or missing)
df = df[df["Meter_Reading"] >= 0]

df = df.dropna(subset=["Meter_Reading"])

# -----

# Generate Summary
# -----

summary = df.groupby("City")["Meter_Reading"].sum().reset_index()

summary.columns = ["City", "Total_Consumption"]

# Save cleaned data
df.to_csv("electricity_cleaned.csv", index=False)

# Save summary
summary.to_csv("electricity_summary.csv", index=False)

print("\nCleaned Dataset:")

print(df)

print("\nElectricity Consumption Summary:")

print(summary)

print("\nFiles generated successfully.")

```

OUTPUT:

File Edit Format Run Options Window Help

```
import pandas as pd
```

```
# Read CSV file
```

```
df = pd.read_csv("electricity.csv")
```

```
print("Original Dataset:")
```

```
print(df)
```

```
# -----
```

```
# Data Cleaning
```

```
# -----
```

```
# Remove rows with missing Consumer ID
```

```
df = df.dropna(subset=["Consumer_ID"])
```

```
# Convert Meter_Reading to numeric
```

```
df["Meter_Reading"] = pd.to_numeric(df["Meter_Reading"], errors="coerce")
```

```
# Remove invalid meter readings (negative or missing)
```

```
df = df[df["Meter_Reading"] >= 0]
```

```
df = df.dropna(subset=["Meter_Reading"])
```

```
# -----
```

```
# Generate Summary
```

```
# -----
```

```
summary = df.groupby("City")["Meter_Reading"].sum().reset_index()
```

```
summary.columns = ["City", "Total_Consumption"]
```

```
# Save cleaned data
```

```
df.to_csv("electricity_cleaned.csv", index=False)
```

```
# Save summary
```

```
summary.to_csv("electricity_summary.csv", index=False)
```

```
print("\nCleaned Dataset:")
```

```
print(df)
```

```
print("\nElectricity Consumption Summary:")
```

```
print(summary)
```

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21)

AMD64) on win32

Enter "help" below or click "Help" above for more information

```
>>>
```

===== RESTART: C:/Users/Hp/OneDrive/Desktop/QUERY/A'

Original Dataset:

	Consumer_ID	City	Meter_Reading
0	1001.0	Chennai	350
1	1002.0	Madurai	420
2	1003.0	Chennai	-50
3	NaN	Salem	250
4	1004.0	Madurai	380

Cleaned Dataset:

	Consumer_ID	City	Meter_Reading
0	1001.0	Chennai	350
1	1002.0	Madurai	420
4	1004.0	Madurai	380

Electricity Consumption Summary:

	City	Total_Consumption
0	Chennai	350
1	Madurai	800

Files generated successfully.

```
>>>
```